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Lab 6 Demodulation and Reception

Introduction

The purpose of this lab was to implement and test a QPSK demodulation and reception system. The lab focused on transmitting a Barker-13 preamble followed by QPSK data, receiving the RF signal using an RTL-SDR, and recovering the transmitted information in MATLAB. Important communication effects such as frequency offset, synchronization, constellation rotation, and bit error rate (BER) were observed and analyzed. This lab demonstrated both the theoretical and practical limitations of digital wireless communication.

Setup

The following equipment was used:

- TM4C123GXL microcontroller
- MCP4822 dual DAC
- RF signal generator with external IQ modulation
- RTL-SDR receiver
- Log-periodic antennas
- MATLAB for signal processing

Transmitter Configuration

The TM4C123GXL was programmed to transmit a Barker-13 BPSK preamble followed by QPSK-modulated data at 8 ksps. The Barker-13 preamble was used because of its strong autocorrelation properties, allowing reliable synchronization at the receiver.

The I and Q baseband signals were generated using the dual DAC and fed into the RF signal generator configured for external IQ modulation at 915 MHz.

Reception and Data Capture

The transmitted signal was captured using the RTL-SDR receiver at a sample rate of 2.048 Msps. One second of IQ data was recorded and stored for MATLAB processing. The captured samples were used for synchronization, demodulation, and BER analysis.

Signal Processing

The IQ data received was processed in MATLAB using several steps:

1. **Frequency Offset Correction**
   A measured frequency offset of -3280 Hz caused constellation rotation in the received signal. This offset was digitally corrected in MATLAB to stabilize the received symbols and remove continuous phase rotation.

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2. **Filtering and Decimation**
   A low-pass FIR filter was applied to reduce out-of-band noise before decimation. The signal was then down sampled to match the 8 ksps symbol rate, allowing symbol decisions to be made at the correct sampling interval.

3. **Preamble Detection**
   Cross-correlation using the Barker-13 sequence was used to locate the preamble and establish timing synchronization. The strong correlation peak identified the start of the transmitted data symbols.

4. **QPSK Demodulation**
   After synchronization, symbols were mapped according to their I/Q quadrant locations to recover transmitted bit pairs. Rotation correction was applied to better align received symbols with ideal QPSK decision boundaries.

5. **Bit Error Rate Measurement**
   Recovered bits were compared with the transmitted data sequence to calculate the bit error rate.

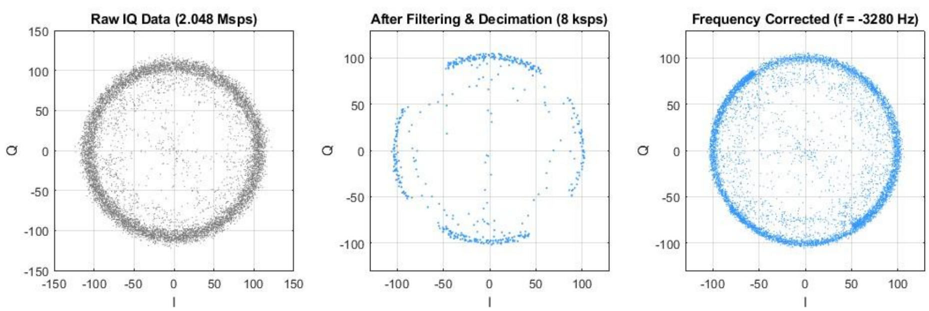

Figure 1. Raw IQ data before processing, after filtering and decimation, and after frequency offset correction.

The raw IQ data at 2.048 Msps shows a circular pattern caused by frequency offset between the transmitter and receiver. After filtering and decimation to the symbol rate, the signal structure becomes much clearer. Following frequency correction, the spinning is removed, and the constellation pattern becomes visible, demonstrating the effectiveness of the correction process.

Results

Parameter	Value
Frequency Offset	-3280 Hz
Symbol Rate	8 ksps
Preamble Type	Barker -13 BPSK (13 symbols)
Correlation Peak	~125,000
Constellation Rotation	~32 degrees
Bit Error Rate (BER)	~45%

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The measured frequency offset caused continuous constellation rotation in the received signal. After filtering, decimation, and frequency correction, the constellation became significantly clearer, and symbol locations were easier to identify.

The BER measured of approximately 43.1% indicates successful synchronization and symbol recovery, but phase rotation introduced some symbol errors. The BER was relatively high, the clear constellation clustering and successful preamble detection verified correct receiver operation.

MATLAB Results:

Frequency Offset: -3280.00 Hz

Preamble Type: Barker-13 BPSK (13 symbols)

Correlation Peak: 125806

Preamble Location: Symbol 16

Data Start: Symbol 29

Best Rotation: 35 degrees

Transmitted Text: "hello hello hello hello hello hello hello hello"

Received Text: "helkÀ††VÆÅ¼ ˆhello ††VÆÆð ˆhello ††VÆÆ"

Bit Error Rate (BER): 43.09%

Preamble detection

Parameter	Value
Preamble	Barker-13
Correlation Peak	125,806
Data Start	Symbol 29

The Barker-13 sequence produced a strong correlation peak which provided reliable synchronization and identified the start of the data symbols.

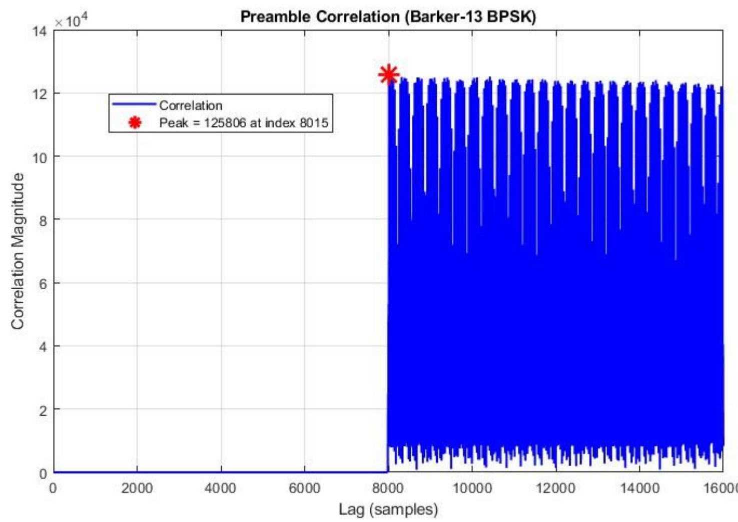

Figure 2. Correlation peak used for preamble detection.

Figure 2 shows a strong correlation peak above 125,000, clearly identifying the start of the Barker-13 preamble. The large peak compared to surrounding values demonstrates reliable synchronization and highlights the strong autocorrelation properties of the Barker sequence.

Constellation Recovery

Parameter	Value
Rotation	80 °
BER	43.1%

The preamble showed expected BPSK clustering while the recovered QPSK data formed four symbol clusters.

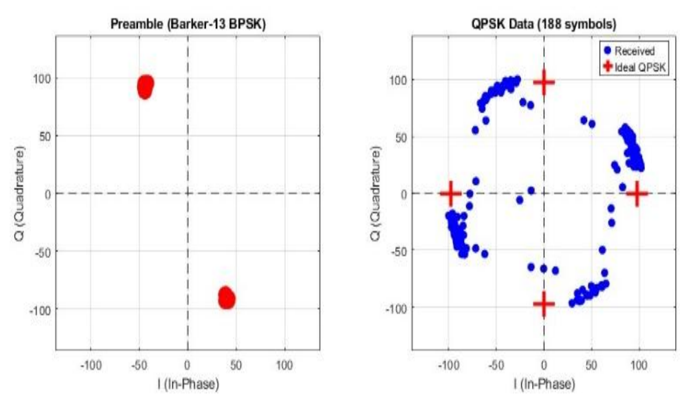

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Figure 3. Preamble constellation and recovered QPSK data.

The preamble constellation shows two distinct BPSK points along the I-axis, while the QPSK data forms four separate symbol clusters. These clusters correspond to the four QPSK symbol locations and confirm successful demodulation, even with some phase rotation and noise present.

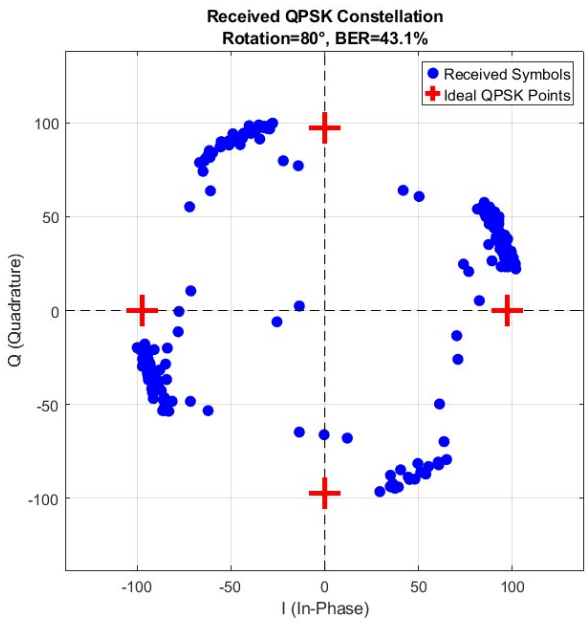

Figure 4. Final received QPSK constellation after rotation correction.

Figure 4 shows the final corrected QPSK constellation with four visible symbol regions. The clusters show some spreading due to phase error and noise, the symbol groupings remain distinguishable and support the measured receiver performance.

Discussion

Frequency Offset Behavior

The measured frequency offset caused continuous constellation rotation and had to be corrected before proper demodulation. This offset resulted from oscillator mismatch between the transmitter and receiver, which is common in practical wireless systems. Even a relatively small frequency offset caused noticeable phase rotation over the capture duration, showing how sensitive coherent demodulation can be to oscillator mismatch. After correction, the signal aligned much better and allowed clearer symbol recovery.

Synchronization Performance

The Barker-13 preamble worked well for synchronization and produced a strong correlation peak much larger than surrounding noise levels. This made it possible to reliably detect the start of transmitted data and establish proper timing for symbol extraction. The strong correlation peak also showed the Barker sequence provided good noise immunity and reliable timing acquisition.

Constellation Recovery

The recovered constellation showed the expected four QPSK symbol clusters, confirming successful demodulation. Some spreading and residual rotation remained due to phase error and noise, the cluster positions still matched expected QPSK behavior.

Bit Error Rate

The BER of about 43% shows synchronization and symbol extraction worked, but residual phase rotation and mapping errors caused symbol decision errors. The BER suggests the system successfully recovered symbols, but improved carrier recovery or finer phase correction could reduce these errors and improve overall receiver performance.

Conclusion

In this lab, a complete QPSK communication system was implemented and tested from transmission through digital demodulation and recovery. Frequency offset correction, preamble detection, filtering, and constellation recovery all operated as expected.

The recovered constellation and strong Barker correlation peak confirmed successful synchronization and symbol extraction, while the measured BER highlighted practical limitations caused by phase rotation and symbol decision errors. These results demonstrated both the theoretical operation of QPSK demodulation and the effects of real-world impairments in wireless systems.

Overall, this lab reinforced important communication concepts, including synchronization, frequency offset compensation, digital filtering, and constellation-based demodulation, while showing how measured results compare with ideal performance.